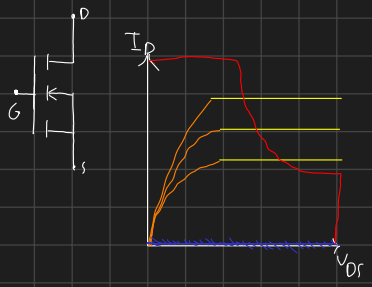
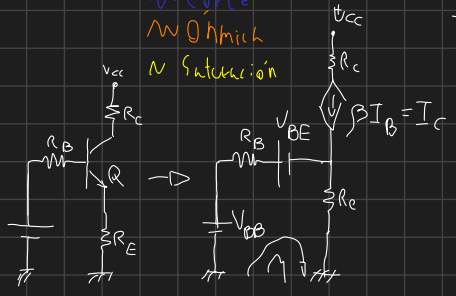
Polarización de transistores (BJT y FET)



- Corte
- Óhmica
- Saturación



$$I_E = I_C + I_B = (\beta+1) I_B$$

$$M \rightarrow -V_{BB} + R_B I_B + V_{BE} + (\beta+1) I_B R_E = 0$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B + R_E(\beta+1)}$$

$$I_C = \beta \cdot I_B = \frac{\beta (V_{BB} - V_{BE})}{R_B + R_E(\beta+1)}$$

$\frac{\Delta V_{BE}}{V_{BE}}$ (5%), $\frac{\Delta \beta}{\beta}$ (3%) $\rightarrow \frac{\Delta I_C}{I_C}$

$$\int_{\beta}^{I_C} = \frac{\beta}{I_C} \cdot \frac{d I_C}{d \beta} = \frac{R_B + R_E(\beta+1)}{(V_{BB} - V_{BE})} \cdot \frac{(V_{BB} - V_{BE})(R_B + (\beta+1) R_E - \beta R_E)}{(R_B + (\beta+1) R_E)^2} =$$

$$\frac{\Delta I_C}{I_C} = \int_{\beta}^{I_C} \frac{\Delta \beta}{\beta} + \int_{V_{BE}}^{I_C} \frac{\Delta V_{BE}}{V_{BE}}$$

$$= \frac{R_B + (\beta+1) R_E - \beta R_E}{R_B + (\beta+1) R_E} = \frac{R_B + R_E}{R_B + (\beta+1) R_E}$$

$$\frac{\Delta I_C}{I_C} = \frac{R_B + R_E}{R_B + (\beta+1) R_E} \cdot 0{,}03$$

$$\frac{\Delta I_C}{I_C} = \frac{R_B + R_E}{R_B + (\beta+1) R_E} \cdot (0{,}03) + \frac{-V_{BE}}{V_{BB} - V_{BE}} \cdot 0{,}05$$